MATH6006: Statistical Methods for OR Modelling

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Continuous Distributions

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Density Functions

A continuous random variable X is described by its **probability density function** (pdf), where

- $f, f \geq 0$
- $\int_{-infty}^{\infty} f(x)dx = 1$

Properties of the PDF

Where $f(x)$ is a pdf

$$P(a \leq X \leq b) = \int_a^b f(x)dx, \quad \text{for } a \leq b$$

Note

- If $a = b$, $P(X = a) = \int_a^a f(x)dx = 0$. That is, for the continuous random variable X, its probability at an individual value is 0. Consequently,
$$P(a \leq X \leq b) = P(a < X < b).$$
- For a continuous RV, X, $P(a \leq X \leq b) = P(a < X < b)$.

Cumulative Distribution Function (CDF)

Definition of CDF

The cumulative distribution function (cdf) is defined as

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- $F(-\infty) = 0$
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Relating the cdf to the pdf

$$F'(x) = f(x)$$

Percentiles, Mean and Variance

- The $(100p^{\text{th}})$ percentile is a number of x_p such that the area under $f(x)$ to the left of x_p is p. That is

$$p = \int_{-\infty}^{x_p} f(x)dx = F(x_p).$$

That is

$$x_p = F^{-1}(p).$$

- The expected value is given by

$$\mu = E[X] = \int_{-\infty}^{\infty} xf(x)dx.$$

- The variance is given by

$$Var(X) = E(X - E[X])^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx.$$

Well-Known Continuous Distributions

We will discuss the following continuous distributions.

1. Uniform distribution
2. Exponential distribution
3. Normal or Gaussian distribution
4. Log-normal distribution
5. Gamma distribution
6. Weibull distribution
7. Beta distribution
8. χ^2 distribution
9. t-distribution
10. F-distribution

Uniform Distribution, $X \sim U(a, b)$

Definition

A continuous random variable X is uniformly distributed if its density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

We write $X \sim U(a, b)$

Mean and Variance

The mean and variance of the uniform distribution are

$$\mu = \frac{a+b}{2}, \qquad \text{and} \qquad \sigma^2 = \frac{(b-a)^2}{12}.$$

Exponential Distribution $X \sim \exp(\lambda)$

- The Poisson distribution is used to examine the number of occurrences of an event over specific time or space interval.
- The exponential distribution gives a description of the length of the time interval between occurrences of the event.
- Hence, the time T until the first event in a Poisson process has an exponential distribution.

Deriving the Exponential Density Function from a Poisson Distribution

- Let $Y_t \sim \texttt{Poisson}(\lambda t)$ be the number of events up to time t.
- Then

$$P(Y_t = 0) = e^{-\lambda t},$$

which means (note: T is the time that the first event occurs! The fact $T > t$ means that no events have occurred between0 and t: $Y_t = 0$)

$$P(T > t) = P(Y_t = 0) = e^{-\lambda t}.$$

- Hence

$$F_T(t) = P(Y \leq t) = 1 - P(T > t) = 1 - e^{-\lambda t}.$$

- The density function of T is

$$f_T(t) = F'(t) = \lambda e^{-\lambda t}.$$

This is the exponential density function.

Exponential Distributions

Definition

A continuous random variable T has an **exponential** distribution if its density function is given with

$$f(t) = \lambda e^{-\lambda t}, \qquad t \geq 0, \qquad \lambda > 0.$$

We write $T \sim \texttt{exp}(\lambda)$.

1. $P(a \leq T \leq b) = \int_a^b \lambda e^{-\lambda t}\ dt = e^{-\lambda a} - e^{-\lambda b}$;
2. $F(t) = P(T \leq t) = \int_0^t \lambda e^{-\lambda x}\ dx = 1 - e^{-\lambda t}$;
3. $f(t) = F'(t) = \lambda e^{-\lambda t}$;
4. Median M is a value for which $F(M) = 1/2$, so $M = (\ln 2)/\lambda$;
5. Mean $\mu = \int_0^\infty t f(t)\ dt = \int_0^\infty t\lambda e^{-\lambda t}\ dt = 1/\lambda$;
6. Variance $\sigma^2 = \int_0^\infty (t-\mu)^2 f(t)\ dt = 1/\lambda^2$;

Memoryless Property of the Exponential Distribution

Memoryless

Let $T \sim \exp(\lambda)$ and $x, y \geq 0$, then

$$P(T > x + y \mid T > x) = P(T > y).$$

Proof.

$$\begin{aligned} P(T > x + y \mid T > x) &= \frac{P(T > x + y,\ T > x)}{P(T > x)} \\ &= \frac{P(T > x + y,\ T > x)}{P(T > x)} \\ &= \frac{e^{-\lambda(x+y)}}{e^{-\lambda y}} = e^{-\lambda y} = P(T > y). \end{aligned}$$

Examples of Exponential Distribution

Most variables that follow an exponential distribution are events that occur at a constant rate λ, where the time until the next event occurs, $T \sim \exp(\lambda)$.

- Times between calls at a telephone exchange,
- Times between two customers at a post office,
- Times between emissions of unstable radioactive elements,
- Time between two accidents at an intersection,
- Lifetime of some type of instrument, etc.

Exponential Distribution: Example

Example

An industrial plant in England has a mean number of lost-time accident per week equal to $\lambda = 0.4$. The number of accidents follows a Poisson distribution. What is the probability that the time between accidents is less than 2 weeks.

Exponential Distribution: Example

Example

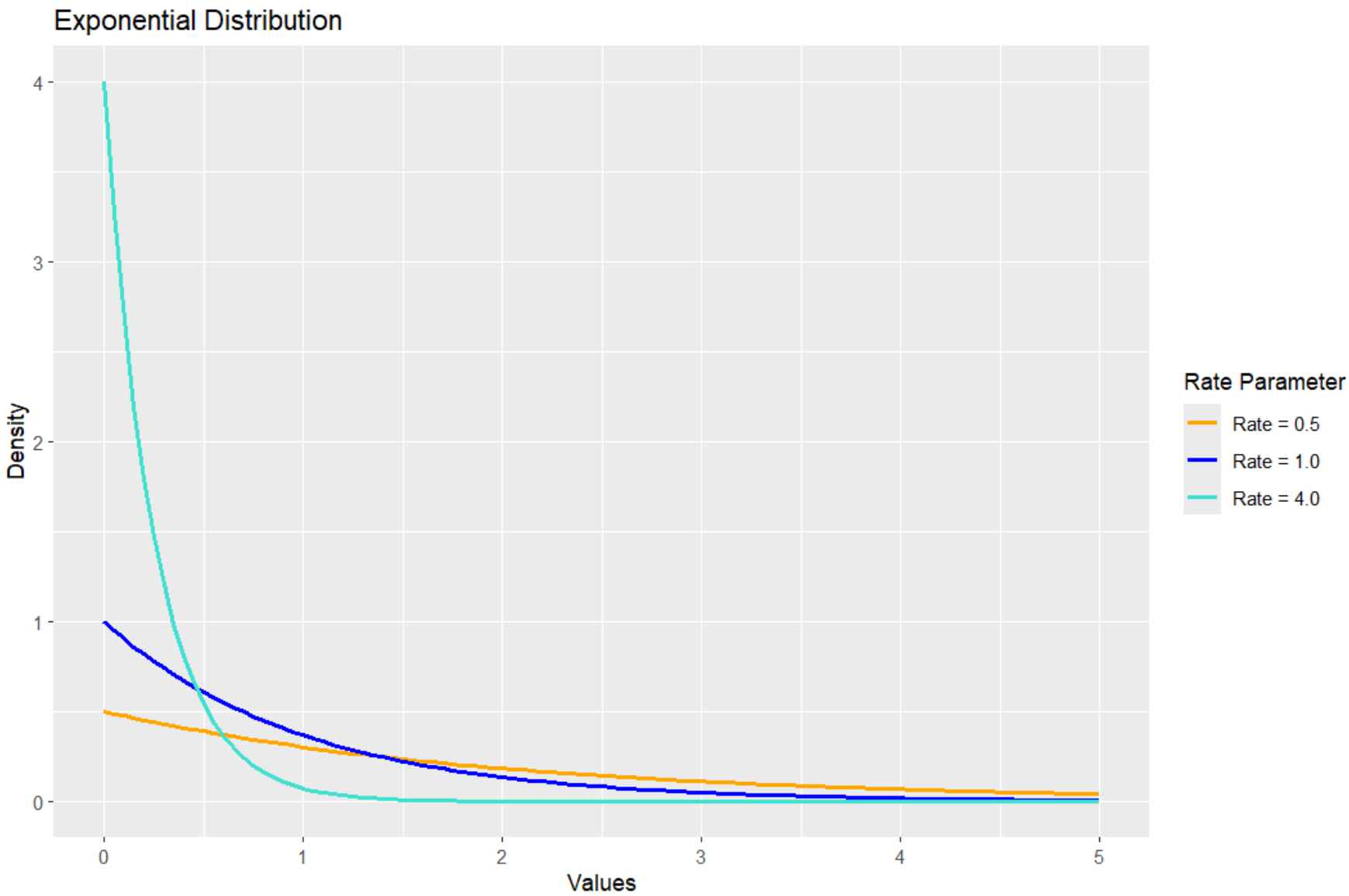
An industrial plant in England has a mean number of lost-time accident per week equal to $\lambda = 0.4$. The number of accidents follows a Poisson distribution. What is the probability that the time between accidents is less than 2 weeks.

Solution: Since the number of accidents follows $\texttt{Poisson}(\lambda)$, then the time T between occurrences of accidents follows the exponential distribution $\texttt{exp}(\lambda)$. It follows that

$$F(T \leq 2) = F(2) = 1 - e^{-2\lambda} = 1 - e^{-0.8} = 0.5507.$$

Thus, the probability of less than 2 weeks between accidents is about 55%.

Exponential Distributions: Different Scales λ



Weibull Distribution: Preliminaries

The Weibull distribution allows the mean number of arrivals to be time dependent, $\lambda(t)$. It is often used to describe the time to failure, where an arrival corresponds to a failure.
Let T be the time until an event occurs, such as a breakdown of some mechanical component. We can think of T as the lifetime of that component.

Survival function

The survival function is defined by

$$S(t) = P(T > t) = 1 - F(t).$$

The survival function $S(t)$ is the probability that the component survives beyond time t.

Weibull Distribution: Preliminaries

Hazard function

The hazard function $\lambda(t)$ is defined by

$$\lambda(t) = \lim_{h \to 0} \frac{P(T \leq t + h \mid T > t)}{h}$$

The hazard function represents the instantaneous failure rate at t per unit of time, given that the system has already survived until time t.

Hazard function, density function and survival function

We have

$$\lambda(t) = \frac{f(t)}{S(t)}.$$

Note: We give no proof of this claim!

Weibull Distributions: From Hazard to Density Functions

From Hazard to Density

Suppose we have the hazard function $\lambda(t)$, then the associated density function is

$$f(t) = \lambda(t)e^{-\int_0^t \lambda(x)dx}.$$

When the Hazard Function is Constant

Exponential distribution revisited

If the hazard function $\lambda(t)$ of a random variable T is constant:

$$\lambda(t) \equiv \lambda,$$

then the density function becomes

$$f(t) = \lambda e^{-\lambda t} \qquad \text{for } t \geq 0.$$

That is $T \sim \exp(\lambda)$.

Note: That the hazard function is constant means that ageing has no effect. That is, the failure of the component does not depend how long it has been used.

Weibull Distributions

The Weibull class of distributions arises from life-time models and are applied to reliability theory. They are the **most popular distributions** used to model the lifetimes of electronic components.

Definition

A random variable T has a Weibull distribution with parameters λ and k if its hazard function is of the form

$$\lambda(t) = k\lambda t^{k-1}, \quad t > 0 \quad \text{for} \quad k,\ \lambda > 0$$

Write $T \sim \texttt{Weibull}(\lambda, k)$.

Note: $\texttt{Weibull}(\lambda, 1)$ is the same as $\texttt{exp}(\lambda)$.

Weibull Distributions: Survival and Density Functions

We have

$$S(t) = e^{-\int_0^t \lambda(x)\ dx} = e^{-\lambda t^k}$$

and

$$f(t) = \lambda(t)e^{-\int_0^t \lambda(x)\ dx} = k\lambda t^{k-1}e^{-\lambda t^k} \text{ for } x \geq 0 \text{ and } \lambda, k > 0.$$

Weibull Distributions: Expectation and Variance

Mean and Variance

$$\mu = \lambda^{-1/k}\Gamma\left(1+\frac{1}{k}\right)$$

$$\sigma^2 = \lambda^{-2/k}\left[\Gamma\left(1+\frac{2}{k}\right) - \left(\Gamma\left(1+\frac{1}{k}\right)\right)^2\right]$$

Here Γ is the gamma function:

$$\Gamma(p) = \int_0^\infty x^{p-1}e^{-x}\,dx \text{ for} p > 0;$$

$$\Gamma(p) = (p-1)\Gamma(p-1) \text{ for all } p > 1; \Gamma(1) = 1; \Gamma(1/2) = \sqrt{\pi};$$

$$\Gamma(n) = (n-1)!.$$

Weibull Distributions: Same Shape with Different Scales

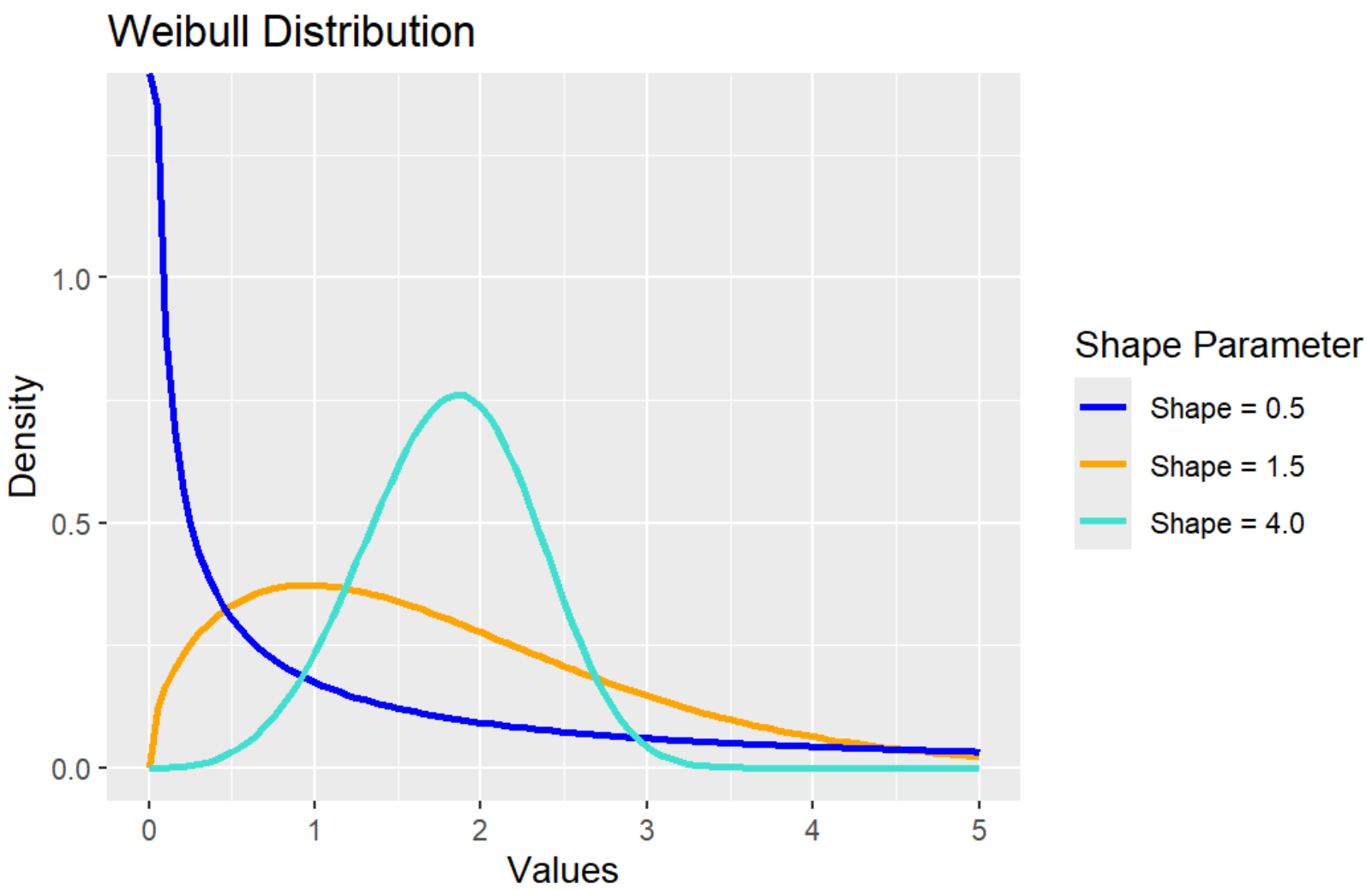


Figure: Shapes of Weibull Distributions

Gamma Distribution

Let X be the waiting time for k random events to occur, occurring at constant rate λ. That is, X is the sum of k independent exponentially distributed random variables with parameter λ,

$$X = \sum_{i=1}^{k} X_i, \quad X_i \sim \exp(\lambda), \ i.i.d.$$

Then the random variable X has a `Gamma` distribution.

Definition via Density Function

A continuous random variable X has a Gamma distribution if its density function is given by

$$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)} \text{ for } x \geq 0.$$

We write $X \sim \Gamma(\lambda, k)$ or $X \sim \texttt{Gamma}(\lambda, k)$.

Gamma Distribution

The definition holds not just for integer values of k but for all positive k. For integer values of k the Gamma distribution is also known as the **Erlang** distribution.

Mean and Variance

$$E[X] = \frac{k}{\lambda}, \text{ and } Var(X) = \frac{k}{\lambda^2}.$$

Proof: It follows from

$$X = \sum_{i=1}^{k} X_i$$

that

$$E[X] = \sum_{i=1}^{k} E[X_i] = \frac{k}{\lambda} \quad \text{and} \quad Var(X) = \sum_{i=1}^{k} Var[X_i] = \frac{k}{\lambda^2}$$

Example: Call Centre

- Telephone calls arrive at an office at a mean rate of $\lambda = 2$ per minute according to a Poisson process. Let X be the waiting time in minutes until the fifth call arrives.
- X can be viewed as

$$X = X_1 + X_2 + X_3 + X_4 + X_5$$

each $X_i \sim \exp(\lambda)$ and they are independent.

- The pdf of X with $k = 5$ and $\lambda = 2$ is

$$f(x) = \frac{2^5 x^4}{4!} e^{-2x},$$

and the mean is $\mu = k/\lambda = 5/2$, the variance is $\sigma^2 = k/\lambda^2 = 5/4$.

Graphs of Gamma Density Functions

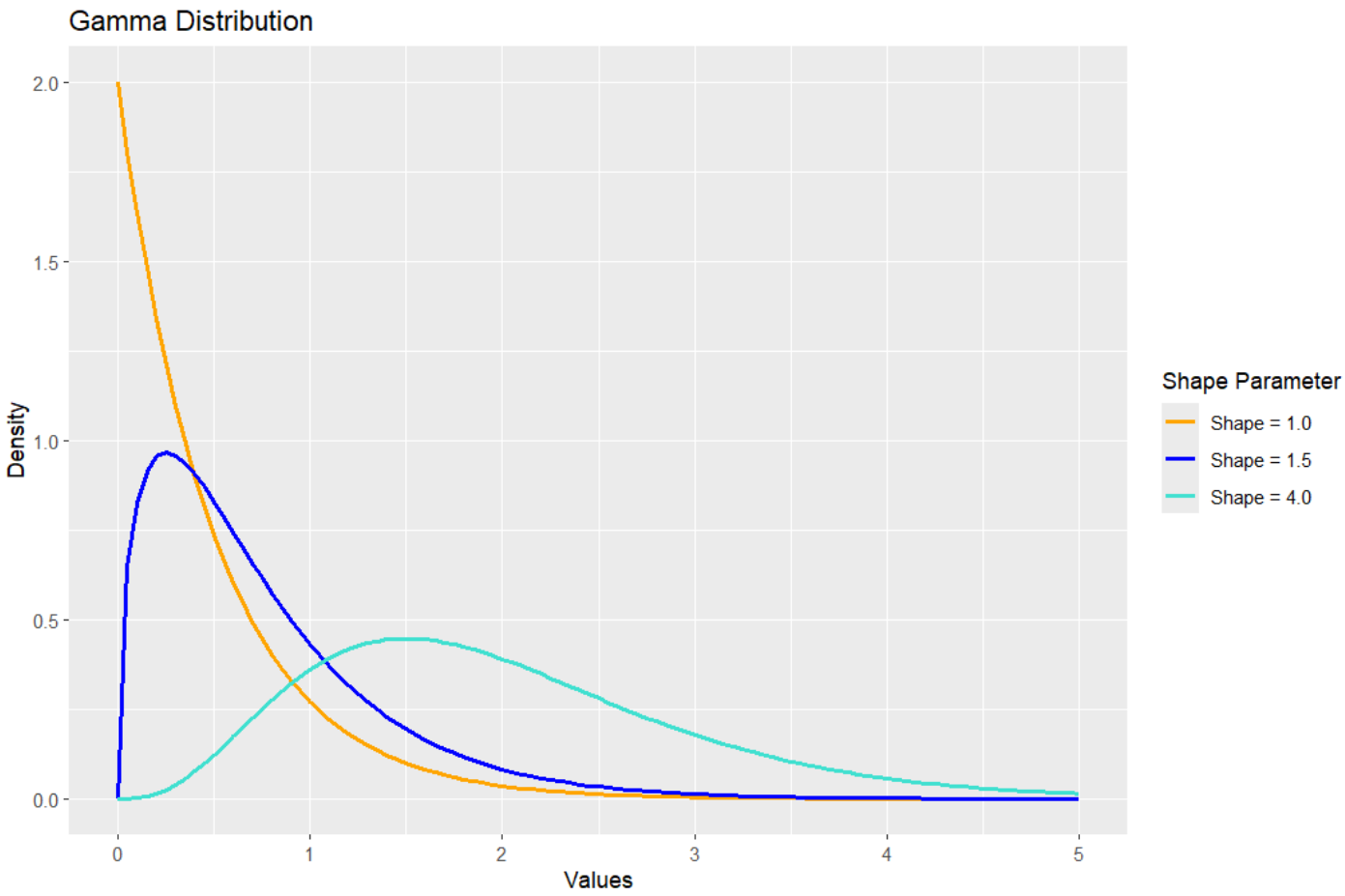


Figure: Shapes of Gamma Distributions

Normal Density Function

A random variable X is said to be normal if its density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

We write $X \sim N(\mu, \sigma)$.

Standard Normal Distributions: $N(0,1)$

Density function

The density function of the standard normal variable Z is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}.$$

From $N(\mu, \sigma)$ to $N(0,1)$

Z-transformation:

$$Z = \frac{X - \mu}{\sigma}.$$

Normal Approximation of Binomial Distribution

We have seen that the number of successes in n independent trials with a constant probability of success p follows a binomial distribution.
Under a binomial distribution, the probability of k successes is

$$P[k] = \left(\begin{array}{c} n \\ k \end{array} \right) p^k (1-p)^k.$$

For large values of n, the binomial distribution can be approximated by the normal distribution with mean np and variance npq.

Lognormal Distributions

A random variable X is said to have a lognormal distribution with parameter μ and σ if its logarithm follows $N(\mu, \sigma)$. That is, $X = e^Y$ with $Y \sim N(\mu, \sigma)$.

Definition

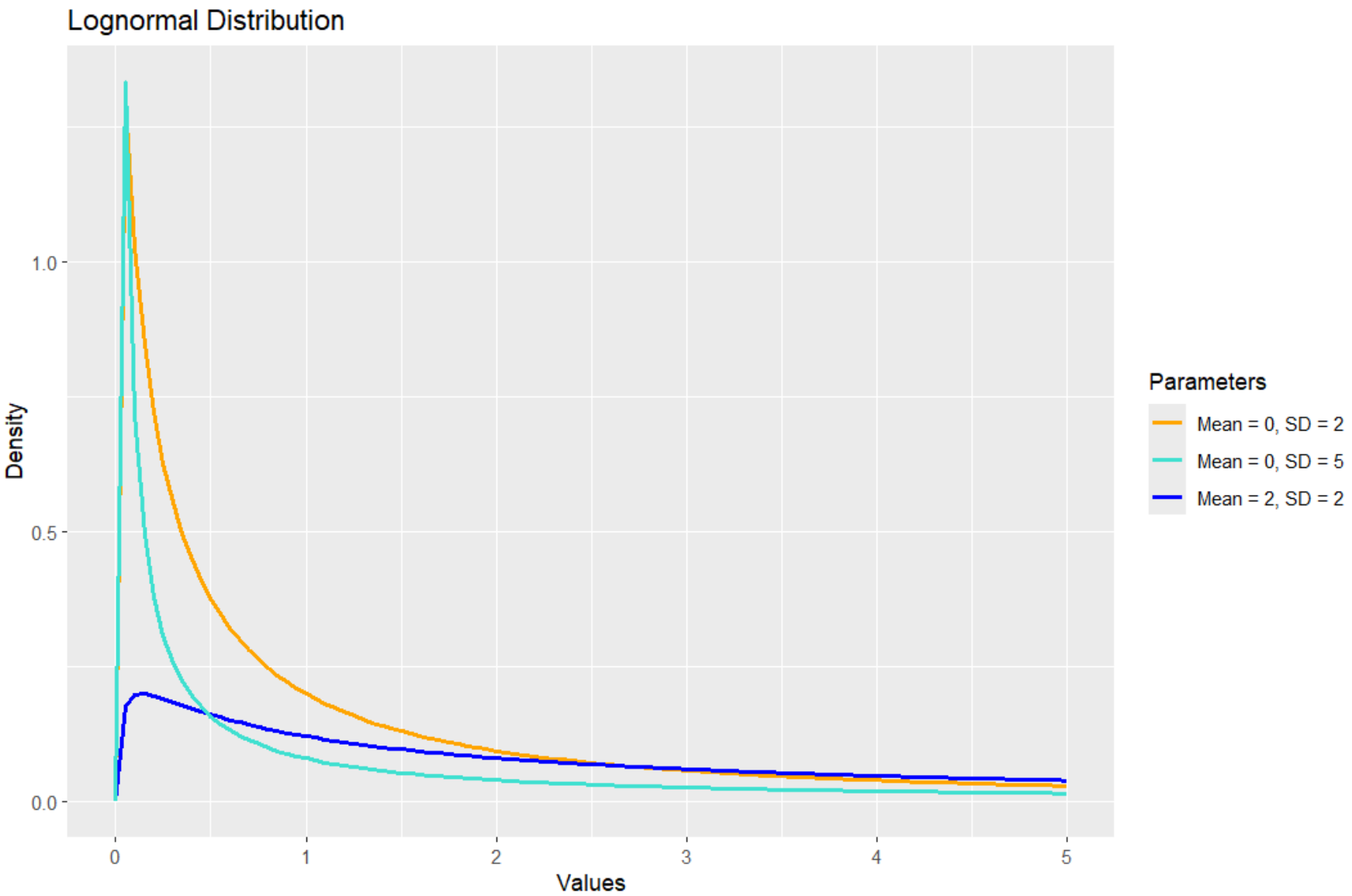
If $Y \sim N(\mu, \sigma^2)$ then $X = e^Y$ is said to have a lognormal distribution. The density of X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}x} exp\{-(\log(x) - \mu)^2/2\sigma^2\} \text{ for } 0 < x < \infty.$$

We write $X \sim \texttt{Lognormal}(\mu, \sigma)$.

Note: Lognormal distribution is extensively used to model a length of service or stock prices but also in many other applications.

Graph of Lognormal Distributions



Lognormal Distributions: Median, Mean and Variance

Median, Mean and Variance

(i) mode is $e^{\mu-\sigma^2}$;

(ii) Median

$$M = e^{\mu}$$

(iii) Mean

$$E(X) = e^{\mu+\frac{\sigma^2}{2}}$$

(iv) Variance

$$Var(X) = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1).$$

Sampling Distributions

The χ^2, t and F distributions are known as **sampling distributions**. They are derived from the normal distribution, and are used when performing hypothesis tests or calculating confidence intervals. For each of these distributions, the statistical tables of percentage points are widely available.

Percentage Point

The point x_α such that

$$P(X > x_\alpha) = \alpha$$

is known as the $100(1-\alpha)\%$ percentage point of a random variable X.

χ^2 Distribution

Definition

Suppose $Z_1, ..., Z_\nu$ are i.i.d. $N(0,1)$, then the random variable

$$X = Z_1^2 + \cdots + Z_\nu^2 \text{ has } \chi^2_\nu \text{ distribution}$$

The probability density function is

$$f(x) = \frac{1}{x^{\nu/2}\Gamma(\frac{\nu}{2})} x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}} \text{ for } x \geq 0,$$

where $\Gamma(\frac{\nu}{2})$ denotes the Gamma function.
We say that X has a chi-squared distribution with ν degrees of freedom, and we write $X \sim \chi^2_\nu$.

χ^2 Distribution: Mean and Variance

Mean and Variance

$$E[X] = \nu \qquad \text{and} \qquad Var(X) = 2\nu.$$

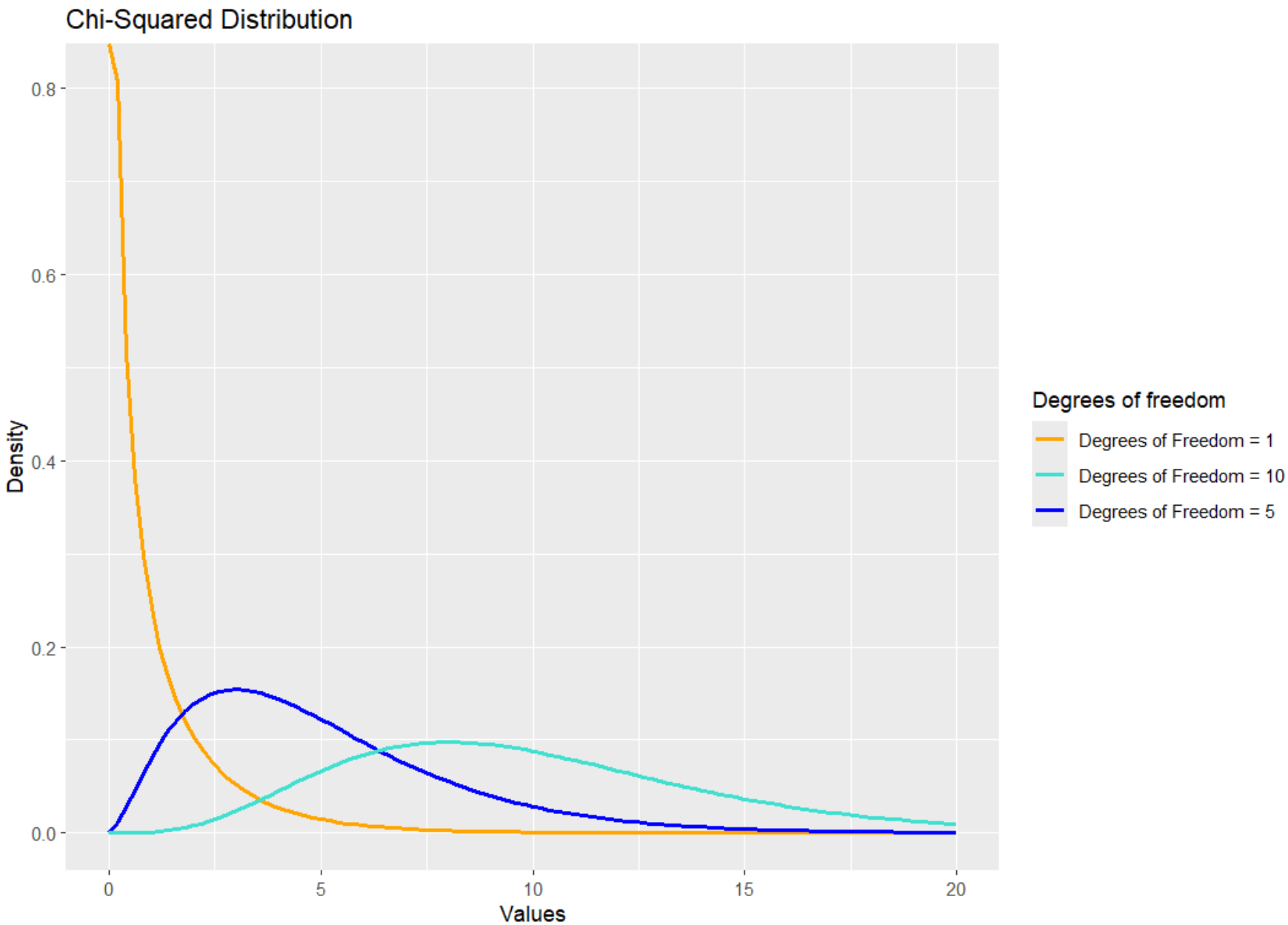
Application: Goodness of Fit

Suppose X_i, $i = 1, 2, \ldots, n$ are i.i.d. random variables with the mean μ. Then $Z_i = X_i - \mu$ represents the error from X_i. Then

$$Z = \sum_{i=1}^{n} Z_i^2 / n$$

is known as the mean of the squared-errors. We often use χ^2 distribution to measure Z.

Graph of χ^2 Distributions



Student's t Distribution

Definition

Let X and Y are independent random variables and $X \sim N(0,1)$, $Y \sim \chi^2_\nu$. Then the random variable

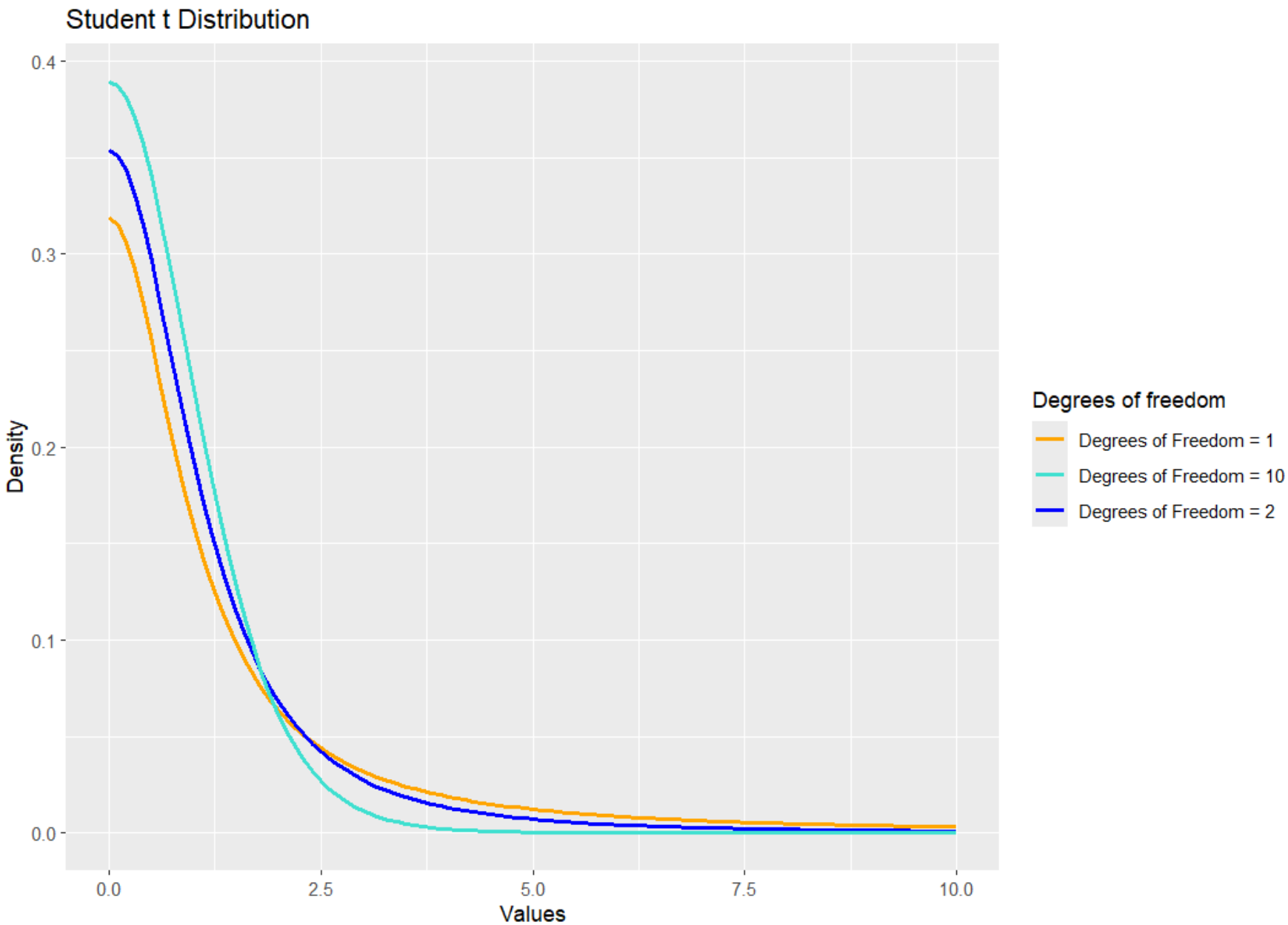
$$T = \frac{X}{\sqrt{Y/\nu}}$$

has a t distribution with ν degrees of freedom. We write $T \sim t_\nu$.
The density of T is given as

$$f(x) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)}\left(1+\frac{x^2}{\nu}\right)^{-(\nu+1)/2} \quad \text{for } -\infty < x < \infty$$

Note: The t_ν distribution is symmetric and its shape is similar to that of $N(0,1)$, but with longer tails. As $\nu \to \infty$, $t_\nu \to N(0,1)$.

Graph of Student's t Distribution



F Distribution

The F distribution is used in ANOVA tests to compare variances.

Definition

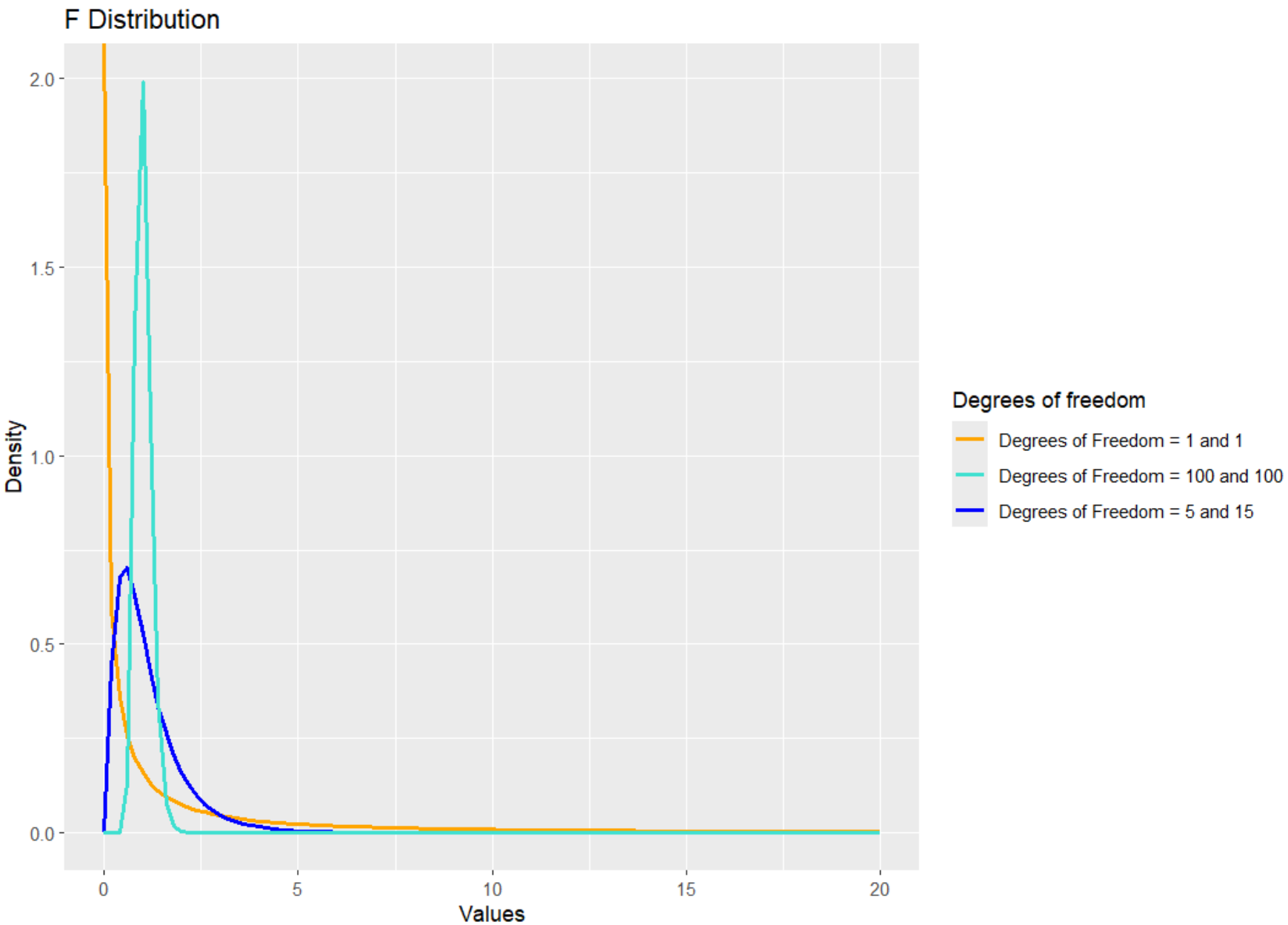
If X and Y are independent random variables with $X \sim \chi^2_\alpha$ and $Y \sim \chi^2_\beta$, then

$$T = \frac{X/\alpha}{Y/\beta}$$

is said to have an $F_{\alpha,\beta}$ distribution.
We write $T \sim F_{\alpha,\beta}$. The density function of T is given as

$$f(x) = \frac{\Gamma((\alpha+\beta)/2)}{\Gamma(\alpha/2)\Gamma(\beta/2)} \left(\frac{\alpha}{\beta}\right)^{\alpha/2} x^{(\alpha/2)-1} \left(1 + \frac{\alpha}{\beta}x\right)^{-(\alpha+\beta)/2}, \text{ for } x > 0.$$

Shape of F Distribution



Distributions of Mixed Type

So far, we have considered random variables that are either discrete or continuous. However, on some occasions, combinations of the two types of random variables are also encountered. That is, in some experiments, positive probability is assigned to each of certain points and also spread over an interval of outcomes.

Example

A fair coin is tossed. If it is Head, the player receives 2. If it is Tail, the player spins a balanced spinner that has a scale from 0 to 1. Then the player receives that fraction of a pound associated with the point selected by the spinner. If X denotes the amount received, what is the probability function of X.

Solution:

There is $1/2$ chance for $x = 2$; and the other $1/2$ chance that the player is going to spin the spinner, which is uniformally distributed over $[0, 1]$. Hence the pdf and cdf are respectively

$$p(x) = \begin{cases} 0 & \text{when } x < 0 \\ 1/2 & \text{when } 0 \leq x \leq 1 \\ 0 & \text{when } 1 < x < 2 \\ 1/2 & \text{when } x = 2 \\ 0 & \text{when } x > 2 \end{cases} \quad \text{and} \quad F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{x}{2} & \text{for } 0 \leq x \leq 1 \\ \frac{1}{2} & \text{for } 1 < x < 2 \\ 1 & \text{for } x \geq 2. \end{cases}$$

The mean is

$$\mu = E[X] = \int_0^1 x/2 dx + 2 \times \frac{1}{2} = \frac{1}{4} + 1 = \frac{5}{4}.$$

The variance is

$$\sigma^2 = E[X^2] - \mu^2 = \int_0^1 x^2/2 dx + 2^2 \frac{1}{2} - (5/4)^2 = \frac{29}{48}.$$